Problem Statement and Goals
Squigglr

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Table 1: Revision History

Date	Developer(s)	Change
26 September 2026	Rayan Nasrallah	Added the Problem Statement section, covering the problem, inputs and outputs, stakeholders, and environment, with supporting references.
26 September 2026	Rayan Nasrallah	Added the reflection responses to the appendix and the generative AI use disclosure.
26 September 2026	Moustafa Moustafa	Added the Goals and Stretch Goals section.
26 September 2026	Moustafa Moustafa	Added my reflection responses to appendix.
26 September 2026	Shamil Canbolat	Added the Custom Documents section
26 September 2026	Shamil Canbolat	Added my reflection responses to appendix
26 September 2026	Ahmed Elzaria	Added my reflection responses to appendix.
26 September 2026	Muhammad Huzaifah	Added my reflection responses to appendix.
28 September 2026	Ahmed Elzaria	Reviewed the document against the course checklists and Development Plan. Clarified the problem, inputs and outputs, stakeholders, and environment; refined goals and completion checks; updated custom document descriptions and AI disclosure; converted references to BibTeX; and improved heading numbering and formatting.
13 October 2026	Rayan Nasrallah	Revised the goal on support and difficulty so that a supporting adult controls progression, and described the domain specialists as de-identified stakeholders, following the specialist interview and the TA meeting of 7 October 2026.
14 October 2026	Rayan Nasrallah	Made the stylus the main drawing input and added finger input as a stretch goal, following the specialist interview.
14 October 2026	Rayan Nasrallah	Stated that the adult chooses child-dependent values, such as pass accuracy, within a range the team sets, following the supervisor meeting of 14 October 2026.
1 November 2026	Muhammad Huzaifah	Addressed peer review: added a paragraph to the Problem

1 Problem Statement

This section describes the need Squigglr addresses, its inputs and outputs, the stakeholders involved, and the intended usage environment.

1.1 Problem

Visual-motor integration is the ability to coordinate visual information with movement. Children rely on it when drawing, copying shapes, and learning to write. Research links these skills to handwriting legibility and identifies motor or handwriting difficulties in some children with attention-deficit/hyperactivity disorder (ADHD) or autism [1, 2, 3]. The need for appropriate practice extends from kindergarten and primary school learners to older children who require continued support. Research on structured foundational activities supports the value of practice for visual-motor development [4].

Children need opportunities to practise at a suitable level and gradually perform activities with less assistance. Parents, caregivers, and educators need to understand both the difficulties encountered and the support needed to complete an activity. A finished drawing alone does not reveal whether a child completed it independently, relied on tracing guides, or corrected difficulties during the attempt. Comparing final drawings without this context can obscure changes in independent performance and make it harder to identify appropriate next steps for practice.

Existing apps meet parts of this need. Handwriting and tracing apps guide children through letters, numbers, and shapes, sometimes with less help at each step; fine-motor practice apps report scores or compare a child with reference ranges; and children's drawing apps offer free drawing. A survey of twelve such products, in the Off-the-Shelf Solutions section of the Software Requirements Specification, found none that combines what this problem requires, and three gaps stand out. None of the products is described as recording which guide a child used, which difficulties occurred, and which were later corrected (G.5). The products that report to adults screen, score, or compare children rather than separating guided attempts from independent ones (G.6). Drawing assistants replace a child's sketch with an artist's drawing or a perfect shape instead of keeping the drawing the child's own (G.4). Graded tracing practice (G.1, G.2) does exist, but mostly in installed apps centred on letters and numbers, and the one browser-based tracing tool found keeps no records. Squigglr aims to offer these parts together.

Squigglr aims to support visual-motor practice through drawing activities with adjustable guidance and difficulty, while making the practice process understandable to supporting adults. Individual attempt records and progress summaries will distinguish independent performance from assisted completion and show recurring difficulties and changes over time. Guided lessons practise lines and shapes on their own; a creative space will let children use the same skills in drawings of their own choosing. The domain specialists we consulted said this free drawing space is valuable and that choice keeps children motivated

to practise. The intended benefit is to help children practise with increasing independence and help supporting adults make informed decisions about further practice. Clinical diagnosis and treatment are outside the project scope.

1.2 Inputs and Outputs

Squigglr accepts activity selections, drawing attempts, and editing actions. It produces drawing guidance, activity feedback, editable artwork, and records that help supporting adults interpret progress.

1.2.1 Inputs

- Selected drawing activities and guidance or difficulty preferences.
- Drawing strokes entered with a stylus on a touchscreen device during practice and creative activities.
- Finished creative drawings, the setting that turns refinement on or off, and editing actions, such as moving, resizing, recolouring, erasing, or undoing changes to drawing elements.
- Requests from authorised adults to view a child's activity history, individual attempts, and progress summaries.

1.2.2 Outputs

- Drawing targets, demonstrations, and guidance appropriate to the selected activity and current support level.
- Activity feedback and completion results based on predefined observations, such as missed tracing points or incomplete lines.
- Suggested changes to guidance or activity difficulty, based on performance, for a supporting adult to accept or decline.
- Editable creative artwork with supported automatic refinements and the ability to restore the original drawing.
- Attempt records showing the activity, guidance used, encountered difficulties, changes during the attempt, and final outcome.
- Progress summaries showing changes across attempts and recurring difficulties, with access to the supporting records.

1.3 Stakeholders

Squigglr's stakeholders include the children using its drawing activities, the adults supporting their practice, and those advising the project.

- **Children:** Kindergarten and primary school learners are the main intended users of the drawing activities. Older children requiring continued support may also benefit from appropriately selected activities. Their interests include understandable guidance, suitable difficulty, and opportunities for independent practice and creative expression.
- **Children with additional support needs:** Children with autism, ADHD, or other developmental or learning differences are included within the intended users. Their individual needs inform the flexibility of guidance, difficulty, and interaction design; a diagnosis alone does not determine the support a child requires.
- **Parents, guardians, and caregivers:** These adults support practice outside school and review activity histories and progress summaries. They need understandable information about assistance used, recurring difficulties, and changes in independent performance.
- **Educators and learning-support staff:** These stakeholders can inform activity selection and interpret practice records alongside their observations of a learner. Their interests include suitable learning activities and useful information for planning further support.
- **School administrators:** If the application is considered for school use, administrators may influence its adoption. Their interests include suitability for the school setting, device availability, and the resources needed to support its use.
- **Domain specialists:** Three specialists, referred to as P1 to P3, hold doctorates in human-computer interaction and related engineering fields and develop software for children with autism, ADHD, and other learning needs. Their interest is that the application's activities, feedback, and progression suit the intended users. They advised the team through a group interview, and their input informs the requirements. They preferred not to be named, so they are de-identified in project documents.
- **Project supervisor:** Dr. Christopher Anand provides guidance on project scope, feasibility, and technical direction.

1.4 Environment

Squigglr is intended for use at home and in educational settings, with adult support available as appropriate to the learner's needs. Children will complete drawing activities through a supported web browser on a compatible touchscreen device, using a compatible stylus. Parents, caregivers, and educators will review activity records and progress summaries through a supported browser on a tablet or computer.

The application will be accessed through the web without requiring a separate application installation. Internet access is expected for normal use, including retrieving activities and saving or reviewing progress. Offline operation

is outside the initial scope. Specific supported browsers and devices will be established through requirements development and compatibility testing.

2 Goals

The following goals describe the outcomes Squigglr aims to support for children and supporting adults. Each goal includes an observable completion check; detailed requirements and test criteria will be developed in later milestones.

G.1 Support structured drawing practice

Provide opportunities to practise visual-motor coordination through activities involving lines, shapes, and simple drawings. Learners should be able to follow an activity, receive understandable feedback, and repeat it for further practice.

This goal is achieved when the agreed lesson catalogue covers these foundational activities and supports the complete practice cycle. The team will verify this by completing and repeating each activity, checking that its instructions, drawing task, and feedback are consistent with its intended purpose.

G.2 Support progression towards independent drawing

Enable learners to practise an activity with progressively less visual guidance, moving from tracing towards drawing without guides. Learners should have opportunities to demonstrate what they can complete independently.

This goal is achieved when the agreed foundational activities support tracing, reduced guidance, and drawing without guides. The team will verify this by completing an activity at each support level and confirming that its records distinguish guided attempts from independent attempts.

G.3 Let supporting adults control support and difficulty

Allow a supporting adult to set the guidance level and activity difficulty for each learner. The application does not change these settings on its own. It may suggest a change, such as more support after repeated difficulty or less support after consistent success, but the change applies only when an adult accepts it.

This goal is achieved when every change to guidance or difficulty comes from an adult's choice or an adult-accepted suggestion. The team will verify this using prepared sequences of successful, unsuccessful, and mixed attempts, checking that the settings stay unchanged until an adult acts and that suggestions follow the agreed criteria. Values that depend on the child, such as how accurate a drawing must be to pass, are chosen by the adult within a range the team sets; the ranges will be defined in the requirements.

G.4 Support editable creative drawing and refinement

Provide a creative space where children can draw freely, using the lines and shapes they practise in guided lessons. When a child finishes a drawing of a supported shape or simple object, Squigglr refines it by adjusting the child's own strokes. The refined artwork should remain editable within the drawing space so that children can continue developing their work and restore their original drawing if desired. Refinement can be turned off, and it never changes guided lesson attempts or their records.

This goal is achieved when users can finish a drawing that is then refined, continue editing the result, and recover the unmodified drawing. The team will verify this using representative drawings from the agreed supported set, checking that refinement preserves editability, that a failed refinement leaves the existing drawing unchanged, and that with refinement turned off the strokes are kept as drawn.

G.5 Preserve meaningful activity records

Help supporting adults understand how a child completed an activity by retaining the attempt's guidance level, encountered difficulties, changes during the attempt, and final outcome. Difficulties will be described through predefined activity observations, such as missed tracing points or incomplete lines.

This goal is achieved when an attempt can be reviewed without losing earlier difficulties or overlooking subsequent corrections. The team will verify this using prepared attempts with known actions, guidance levels, and outcomes, checking that the records accurately represent both the process and the final result.

G.6 Make progress understandable to supporting adults

Help parents, guardians, caregivers, and educators recognise changes in a child's performance and identify recurring difficulties that may inform further practice. Summaries should distinguish changes in independent performance from results achieved with additional support.

This goal is achieved when an authorised adult can review activity history, understand changes in performance and assistance used, and inspect the attempts supporting each summary. The team will verify accuracy using prepared histories showing improvement, recurring difficulties, and changing support levels. Adult reviewers will also be asked to interpret example summaries and locate supporting attempts to assess whether the information is understandable.

3 Stretch Goals

The following extensions will be considered after the core goals are met, subject to remaining time and technical feasibility.

Camera-based finger tracing.

Provide an alternative interaction in which children trace displayed guides by moving a finger within a live camera view. This extension would be achieved when the application can track the finger's movement relative to a guide and provide activity feedback. The team would verify this using representative tracing attempts under defined camera and lighting conditions.

Letter tracing and drawing.

Extend structured practice to an agreed set of letters, supporting progression from tracing letter outlines to forming letters without guides. This extension would be achieved when the selected letter activities support guided and independent attempts, activity feedback, and attempt recording. The team would verify each activity at the supported guidance levels.

Finger input.

Allow children to draw on the touchscreen with a finger, as an easier alternative to the stylus for children who find holding a tool difficult. This extension would be achieved when structured and creative activities accept finger input as well as stylus input. The team would verify this using representative finger and stylus attempts in each activity.

4 Custom Documents (CDs)

The team proposes a Specialist Interview Report for the fall term and a Usability Report for the winter term, consistent with the Development Plan. These choices remain subject to instructor approval.

Fall — Specialist Interview Report

The report will cover the interview we held with domain specialists. It will state the interview method and questions, identify the participants by code only, include the de-identified transcript as an appendix, and analyze which findings shaped which requirements and hazards.

Winter — Usability Report

The report will evaluate the usability of the drawing activities and adult-facing progress summaries. It will document evaluation tasks, methods, findings, limitations, and recommended improvements. Evaluation will include team walkthroughs and task-based sessions with adult reviewers, including checks of whether reviewers can interpret summaries and locate supporting attempt records.

The evaluation will not depend on child participation. Findings from adult reviewers will inform design improvements, but will not be presented as direct evidence of usability for children or of developmental benefits.

Appendix — Reflection

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?

   Rayan: Through my parents, I connected with a specialist who holds a doctorate and develops software for children with autism, as well as the head of a nonprofit organisation serving people with additional support needs, including autism and ADHD. These connections gave us a clearer direction for identifying stakeholders who could inform our requirements. They also helped me recognise the value of seeking both software-design expertise and experience supporting our intended users. We now have potential contacts who can help us question our assumptions and understand needs that we might otherwise overlook.

   Ahmed: What went well was developing a clearer connection between our users' needs and the goals of the application. I found the distinction between completing a drawing and completing it independently particularly useful. A finished drawing can look successful without showing how much guidance the child needed. Recording assistance and encountered difficulties therefore gives the progress summaries a clearer purpose than simply displaying completed artwork.

   Working on the Development Plan alongside this document also helped me consider whether our proposed approach supported the stated goals. For example, automatic refinement should leave children with artwork they can continue editing, which affects what we need to investigate in the proof of concept. I learned that dividing the documents between team members does not remove the need to check their consistency. For future milestones, I would make this comparison an explicit review step so that our technical plans continue to address the agreed user needs.

   Moustafa: After the first tutorial in the second week, Rayan and I discussed two project ideas with our TA, Chris Schankula. Chris provided valuable insights into what we should consider and how we could approach the drawing application. He also recommended Dr. Anand as a potential

supervisor. This discussion helped us evaluate the idea more clearly and gave us the confidence to pursue it. Having an early opportunity to discuss the project with someone knowledgeable helped establish a direction before we committed further effort to the proposal.

Muhammad: Our team connected easily around the high-level idea and the problem we wanted to solve. This shared understanding helped us align on the problem statement and goals, because we had a common purpose to refer back to when discussing what belonged in the document. I found that agreeing on the problem first made it easier to discuss goals without getting drawn too early into implementation details. My main takeaway is that time spent establishing a shared direction makes later decisions more focused and helps individual contributions fit together.

Shamil: The problem we chose to address already had research behind it with clearly defined stakeholders, namely primary school teachers and children/students. Our initial notes on the topic will translate easily into the literature review custom document. Our supervisor and TA both have background knowledge in education and thus directed us to clear and novel goals. Lastly, our team is excited about the project, so we successfully assigned tasks, built a workflow, and set up regular meetings with little conflict.

2. What pain points did you experience during this deliverable, and how did you resolve them?

   Rayan: When I first presented our idea to Dr. Smith, he explained that we needed to justify the need for the application. I realised that describing useful features was not enough; we needed evidence supporting the problem we wanted to address. I reviewed research on visual-motor development, drawing practice, and difficulties experienced by some children with autism or ADHD. Comparing these findings helped me connect our proposal to existing work and explain the rationale for structured drawing practice. It also clarified the distinction between evidence supporting our approach and evidence that our particular application will be effective. In future deliverables, I would establish the evidence and its limitations earlier so that they guide our scope from the beginning.

   Ahmed: A challenge for me was separating an interesting technical feature from a convincing reason to build it. Our earlier idea included drawing in the air using a camera, but the professor's feedback challenged whether that interaction addressed a meaningful need. This made me reconsider how easily enthusiasm for an implementation can influence the way we describe the problem.

   We responded by focusing the core experience on touchscreen or stylus drawing and keeping camera-based interaction as a stretch goal. Our supervisor's recommendation to develop a web application also gave us

a more focused implementation direction. However, choosing a practical platform does not establish that the application will benefit its users. I see the research supporting the problem and future stakeholder feedback as necessary complements to technical feasibility.

Another point that needed clarification was what automatic refinement would provide. Returning a separate enhanced image would not meet our intention for children to continue editing their drawings. Clarifying this distinction helped connect the goal to an observable result. In future, I would describe what a user must be able to do after each proposed feature before discussing the technology used to build it.

Moustafa: We initially struggled to write goals at the right level. After discussing them with Chris, we concluded that they were too broad, and he emphasised that they needed to be measurable. Our first response was to add numerical targets to the existing goals, but this felt forced and introduced evaluation expectations that depended on testing with children. We then reconsidered what users should actually be able to accomplish and wrote new goals with observable completion criteria that our team could verify. This helped me understand that a measurable goal does not necessarily require a number to be quantifiable; it needs a clear way to establish whether the intended outcome has been delivered. I would apply that distinction earlier when defining goals for future work.

Muhammad: We had minor disagreements about commit and pull request structures and meeting times. Although we agreed on the project idea, these discussions showed me that we still had different expectations about how to work together. We resolved them by giving everyone an opportunity to explain their preferences and constraints, then choosing an approach that the majority agreed on. Hearing the reasons behind each view helped us reach a practical decision. I learned that agreeing on working conventions early is worth the discussion, because small differences can otherwise create repeated friction during shared work.

Shamil: Figuring out where to start with Gitlab/Avenue (checklists and rubrics) and Github (issues, version control, and templates) was confusing at first, but after coordinating online and in person and asking our TA during our meeting, our team was on the same page. Also, attendance and participation was not the best, but we outlined strict and detailed rules in the team charter which our members enforced for productivity.

3. How did you and your team adjust the scope of your goals to ensure they are suitable for a Capstone project (not overly ambitious but also of appropriate complexity for a senior design project)?

   Team response: We narrowed the core scope to structured drawing practice, progression towards independent drawing, adjustable support, free drawing with refinement, attempt recording, and parent-facing progress

summaries. We moved camera-based tracing and letter activities to stretch goals so that they would not distract from completing the core experience. We also limited activity feedback to predefined observations, such as missed tracing points, rather than personalised generated advice.

We distinguished user goals from supporting requirements such as account creation and browser compatibility. We also separated recording individual attempts from summarising progress, giving each goal a distinct purpose. Our completion checks now use prepared drawing attempts and activity histories that the team can evaluate without depending on child participants. These decisions keep the project manageable while retaining substantial design work in drawing assessment, adaptive guidance, editable artwork, and accurate progress reporting.

Generative AI Use Disclosure

Our team used OpenAI GPT-6 Astra to improve the presentation of this document. We independently developed the project ideas and wrote the initial content before using AI. Its role was limited to LaTeX formatting, citation formatting, grammar correction, and improvements to sentence structure, clarity, and flow. No additional tools or skills were used.

All project ideas and substantive decisions originated with the team. We reviewed the suggested edits against our original draft to ensure that they preserved our intended meaning and checked the formatted citations against the source materials.

References

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[3] C. T. Fuentes, S. H. Mostofsky, and A. J. Bastian. Children with autism show specific handwriting impairments. *Neurology*, 73(19):1532–1537, 2009.

[4] Livia Taverna, Marta Tremolada, Liliana Dozza, Renata Zanin Scaratti, Ulrike Domahs, Cristina Lallo, and Barbara Tosetto. Who benefits from an intervention program on foundational skills for handwriting addressed to kindergarten children and first graders? *International Journal of Environmental Research and Public Health*, 17(6):2166, 2020.